Capstone Project:

Data Analytics and Data Science

About Dataset:

The e-commerce platforms analysed in this project refer to major online retail companies operating globally. Data was collected from Companies Market Cap, which ranks companies based on financial indicators as market capitalization, revenue.

This data helps evaluate the performance and scale of leading e-commerce businesses.

Google Colab:

Installing Required Libraries in Colab

In Google Colab, required libraries were installed using:

```
!pip install requests
!pip install beautifulsoup4
!pip install pandas
```

Web Scraping Using BeautifulSoup

The revenue ranking page was scraped using Python in Colab.

Process:

- Sent HTTP request using requests
- Parsed webpage using BeautifulSoup
- Extracted:
  - Company Name
  - Revenue (TTM)
  - Country
  - Rank
- Stored extracted data in a Pandas DataFrame

Saving CSV File in Google Colab

After creating the DataFrame, the dataset was saved using:

```
df.to_csv("ecommerce_revenue.csv", index=False)
```

In Google Colab, files are stored temporarily in:

/content/

So the file path is:

/content/ecommerce_revenue.csv

Downloading CSV File from Colab

To download the file to local system:

```
from google.colab import files
files.download("ecommerce_revenue.csv")
```

This downloads the file directly to the computer.

Data Cleaning in Colab

Data cleaning was performed using Pandas:

- Removed $ and B symbols
- Converted revenue to numeric format
- Checked for null values
- Standardized column names

Exploratory Data Analysis in Colab

Performed:

- Mean Revenue calculation
- Median Revenue calculation
- Top companies identification
- Revenue distribution visualization

Chart were created using:

```
import matplotlib.pyplot as plt
```

Data Type Conversion

- Converted columns to appropriate data types (e.g., price to numeric, ratings to float, review count to integer).
- Removed non-numeric symbols such as currency signs and commas before conversion.

Standardization and Formatting

- Standardized text fields (e.g.,names and categories) by trimming spaces and correcting inconsistent capitalization.
- Ensured uniform units and formats across the dataset.

Outlier Detection

- Identified extreme values using statistical methods and visualization.
- Verified whether outliers were valid or errors, and treated them accordingly.

Feature Preparation

- Created or refined variables required for analysis.
- Ensured all fields were clean, consistent, and analysis-ready.

Tools Used

- Python
- pandas for data manipulation
- NumPy for numerical operations

Outcome

After cleaning, the dataset became structured, consistent, and free from major errors, making it suitable for exploratory data analysis (EDA) and statistical modeling. The preprocessing steps improved data quality, reliability, and overall analytical accuracy.

Advantages of Using Google Colab

Cloud-based execution
No dependency issues
Easy file sharing
Direct CSV download
Ideal for academic projects

Conclusion

The entire revenue-based e-commerce analysis was successfully completed using Google Colab, covering:

- Web scraping
- Data cleaning
- Statistical analysis
- CSV export

Sql:

Data Import Process:

- The cleaned CSV file generated during preprocessing was imported into the SQL database.
- SQL import with MySQL Workbench were used to load the data into the table.
- Data integrity was verified after import to ensure all records were correctly transferred.

Queries:

Retrieve Company Names:

```
SELECT Name
FROM `data`.`ecommerce_revenue`;
```

Explanation:

- This query selects the Name column from the table ecommerce_revenue.
- The table is located inside the database named data.
- It returns a list of all company or platform names stored in the dataset.

Identifying High-Value Platforms by Country

```
SELECT
  Country,
  AVG(Revenue) as Average_Revenue,
  SUM(Revenue) as Total_Revenue,
  COUNT(*) as Company_Count
FROM `data`.ecommerce_revenue
GROUP BY Country
HAVING Company_Count > 5
ORDER BY Total_Revenue DESC;
```

Explanation:

This query analyzes revenue performance by country.

Step-by-step meaning:

- Country → Groups results based on each country
- AVG(Revenue) → Calculates the average revenue per country
- SUM(Revenue) → Calculates the total revenue generated in each country
- COUNT(*) → Counts how many companies or records exist per country
- GROUP BY Country → Combines all rows belonging to the same country
- HAVING Company_Count > 5 → Filters only countries with more than 5 companies
- ORDER BY Total_Revenue DESC → Sorts countries from highest to lowest revenue

Total Revenue and Average Order Value by Country

```
SELECT
  Country,
  SUM(Revenue) as Total_Revenue,
  AVG(Revenue) as Average_Order_Value,
  COUNT(*) as Order_Count
FROM `data`.ecommerce_revenue
GROUP BY Country
ORDER BY Total_Revenue DESC;
```

Explanation:

This query provides an overall revenue summary by country.

Year-over-Year (YoY) Revenue Growth Analysis:

This SQL query calculates how revenue changes from one year to the next for each country.

Yearly Revenue Summary (CTE)

```
WITH YearlyRevenue AS (
  SELECT
    Country,
    EXTRACT(YEAR FROM Order_Date) as Sales_Year,
    SUM(Revenue) as Annual_Revenue
  FROM `data`.ecommerce_revenue
  GROUP BY Country, Sales_Year
)
```

Explanation:

- EXTRACT(YEAR FROM Order_Date) → Gets the year from each order date
- SUM(Revenue) → Calculates total revenue for that year
- GROUP BY Country, Sales_Year → Aggregates revenue per country per year

Retrieve Previous Year Revenue

```
LAG(Annual_Revenue) OVER (PARTITION BY Country ORDER BY Sales_Year)
```

Explanation:

- LAG() gets the revenue from the previous row (previous year)
- PARTITION BY Country → Calculates separately for each country
- ORDER BY Sales_Year → Ensures chronological comparison

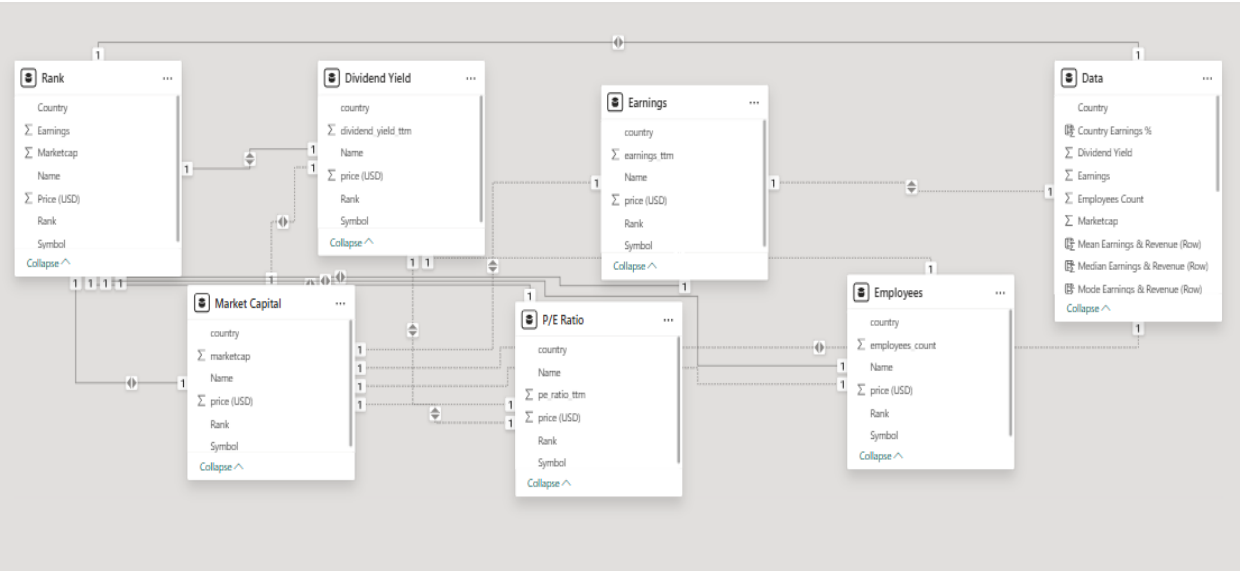
Calculate Growth Percentage:

ROUND(((Annual_Revenue - LAG(Annual_Revenue) OVER (PARTITION BY Country ORDER BY Sales_Year)) /
NULLIF(LAG(Annual_Revenue) OVER (PARTITION BY Country ORDER BY Sales_Year), 0)) * 100, 2)

Explanation:

This formula computes Year-over-Year growth

Power Bi:



Data Modeling in Power BI:

- The cleaned dataset was imported into Power BI and organized into multiple related tables such as Rank, Earnings, Market Capitalization, P/E Ratio, Dividend Yield, Employees, and Country Data. Relationships were established between these tables using common fields (e.g., Name, Symbol, Country) to create a structured data model.
- This model enables efficient analysis, aggregation, and visualization of key business metrics including revenue, earnings, market value, and employee count. Proper relationships ensure accurate filtering, improved performance, and meaningful insights across different dimensions such as country and company.

Outcome:

The data model supports interactive dashboards and advanced analytical reporting.

Dashboard:

Sum of Market cap (Gauge) – Shows the total combined market capitalization of all companies.

Sum of Employees (Card) – Displays the total number of employees across companies

Sum of Earnings (Card) – Shows total earnings generated by all companies

Sum of P/E Ratio by Country (Card) – Displays the total P/E ratio aggregated by selected countries

Sum of Earnings by Country (Table) – Lists earnings values for each country with a total at the bottom.

Sum of Revenue (Gauge) – Shows the total revenue generated by all companies.

Sum of Price (USD) by Country (Card) – Displays the total stock price value aggregated by country

Country, Rank (Slicer Panel) – Allows filtering the dashboard visuals based on selected countries.

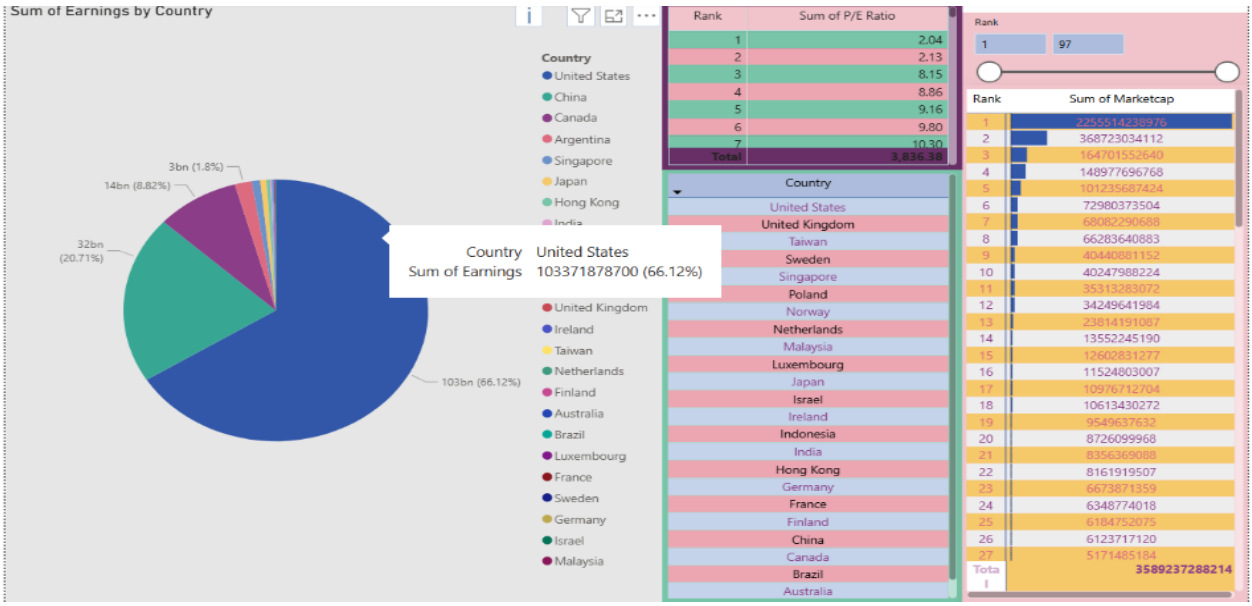
Revenue by Name (Pie Chart) – Shows revenue contribution percentage of each company.

Market cap by Symbol (Donut Chart) – Displays market capitalization distribution by company symbol.

Price (USD) by Country (Stacked Column Chart) – Compares stock prices across different countries.

Revenue and Sum of Earnings (Line and Clustered Column Chart) – Compares total revenue and earnings side-by-side.

2ND Dashboard:



Sum of Earnings by Country (Pie Chart) – Displays each country's percentage contribution to the total earnings.

Rank vs Sum of P/E Ratio (Table) – Shows companies ranked along with their corresponding P/E ratio values.

Total P/E Ratio (Bottom of Table) – Displays the overall sum of P/E ratios for all ranked companies.

Country (Slicer List) – Enables filtering of all visuals based on selected countries.

Rank (Range Slicer) – Allows filtering data dynamically by selecting a specific rank range (1–97).

Rank vs Sum of Market cap (Table with Data Bars) – Displays company ranks with their market capitalization using visual comparison bars.

Total Market cap (Bottom of Table) – Shows the total market capitalization of all selected ranked companies.

Dax Query:

Country Revenue % → shows the share of total revenue generated by companies from a particular country.

Country Earnings % → shows the share of total earnings (net income/profit) contributed by those same companies.

Mean (average) → gives the central tendency by summing revenue and earnings values for that row and dividing by count.

Median → shows the middle value, which is less affected by extreme highs or lows.

Mode → highlights the most frequently occurring value in that row's revenue/earnings distribution.

For Power Bi

Overall Links Used For Web Scraping

1. https://companiesmarketcap.com/e-commerce/largest-e-commerce-companies-by-market-cap/
2. https://companiesmarketcap.com/e-commerce/most-profitable-e-commerce-companies/
3. https://companiesmarketcap.com/e-commerce/largest-e-commerce-companies-by-revenue/
4. https://companiesmarketcap.com/e-commerce/e-commerce-companies-ranked-by-pe-ratio/
5. https://companiesmarketcap.com/e-commerce/e-commerce-companies-ranked-by-dividend-yield/
6. https://companiesmarketcap.com/e-commerce/e-commerce-companies-ranked-by-operating-margin/
7. https://companiesmarketcap.com/e-commerce/largest-e-commerce-companies-by-number-of-employees/

Each Link Represents:

I scraped different rankings of **e-commerce** companies based on **business metrics**.

Largest by Market Capitalization:

Company value in the stock market

Shows how big and valuable the company is to investors

Most Profitable E-commerce Companies

Companies earning the highest profits

Indicates financial success and efficiency

Largest by Revenue

Total sales generated

Shows business scale and sales power

Ranked by P/E Ratio

P/E Ratio=Earnings per Share (EPS)Market Price per Share

Price-to-Earnings ratio

Used by investors to judge stock valuation

Example:

If a company's share price = ₹500
Earnings per share (EPS) = ₹25

P/E = 500 ÷ 25 = **20**

This means investors pay ₹20 for every ₹1 of earnings.

Interpret:

✓ **High P/E** → Investors expect high future growth
✓ **Low P/E** → Company may be undervalued or slow growth.

Ranked by Dividend Yield

Returns paid to shareholders

Shows companies that distribute profits

Ranked by Operating Margin

Profit after operating costs

Measures operational efficiency

Largest by Number of Employees

Workforce size

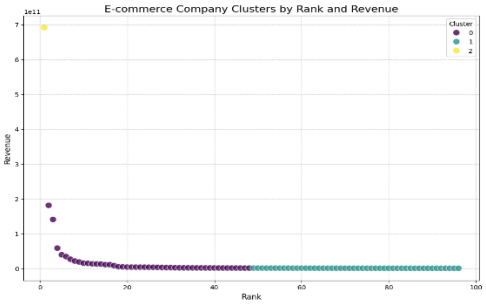
Indicates organizational scale

Data Science

Unsupervised Learning:

K Means:

K-Means is an **unsupervised machine learning algorithm** used for clustering data points into distinct groups based on their inherent similarities. Unlike supervised learning, it works with **unlabeled data** and aims to uncover hidden patterns or structures within the dataset.



The result I got from above showing diagram for k-means clustering is **positive** in the sense that the algorithm has successfully grouped the companies into distinct clusters based on rank and revenue.

Signs of a Positive Result

- **Clear separation of clusters**: The three colors (Cluster 0, 1, and 2) show that k-means has identified meaningful groupings.
- **For example**, Cluster 2 isolates the high-revenue outlier, while Clus ter 0 and 1 capture the bulk of lower-revenue companies.
- **Market segmentation insight**: We can interpret Cluster 2 as "market leaders," Cluster 1 as "mid-tier," and Cluster 0 as "long tail" companies. That's exactly the kind of insight k-means is meant to provide.
- **Outlier handling**: The algorithm didn't force the extreme revenue company into a low-revenue cluster; instead, it gave it its own grouping, which is appropriate.

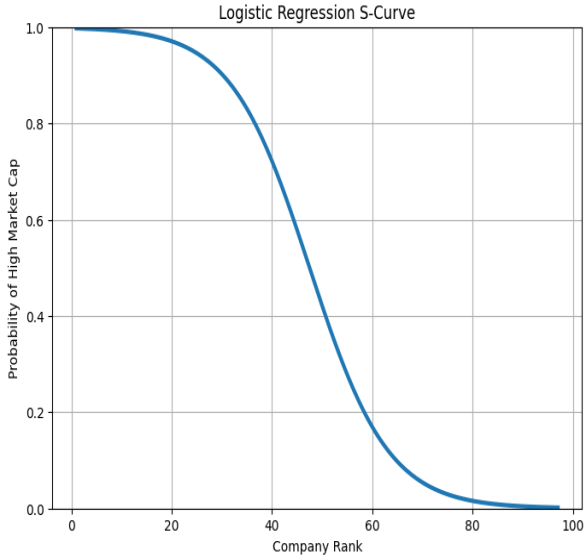
Supervised Learning:

Comparison Table

Feature	K-Means (Clustering)	Logistic Regression
Goal	Grouping similar items	Predicting a label/category
Output	Cluster ID (0, 1, 2)	Probability (0% to 100%)
Input	Unlabeled data	Labeled data (Target exists)

Logistic Regression:

In Python, we need to import it from the sklearn.linear_model module. Logistic Regression is a statistical method for predicting binary classes and is widely used in machine learning for binary classification tasks.



This logistic regression S-curve is showing how the probability of a company having **high market capitalization** changes with its **rank**.

Key Elements of the Curve

- **Sigmoid shape (S-curve)**: Logistic regression always produces this kind of curve because it maps values to probabilities between 0 and 1.
- **X-axis (Company Rank)**: Lower ranks (closer to 0) represent top-performing companies, while higher ranks (closer to 100) represent weaker ones.
- **Y-axis (Probability of High Market Cap)**: This is the predicted probability that a company belongs to the "high market cap" category.

Interpretation

- **Left side (low rank)**: Probability is close to 1.0 → Top-ranked companies are very likely to have high market capitalization.
- **Middle ranks**: The probability gradually decreases → Mid-ranked companies have a mixed chance of being high market cap.
- **Right side (high rank)**: Probability approaches 0.0 → Lower-ranked companies are very unlikely to have high market capitalization.

Why Logistic Regression Fits Here

- It's ideal for **binary classification** (high vs. not high market cap).
- The S-curve captures the **nonlinear relationship**: small changes in rank at the middle range can sharply change the probability, while at the extremes (very high or very low rank), the probability stabilizes.

This curve is a **positive result**: it shows that logistic regression is successfully modeling the relationship between rank and the likelihood of high market cap, giving a clear probability-based interpretation rather than a hard cutoff.

Based on this specific plot, the model has "learned":

1. **Confidence at the Top:** If a company is ranked between **1 and 20**, the probability is nearly **1.0**. The model is extremely confident these are "High Market Cap" companies.
2. **The "Turning Point":** Look at **Rank 45-50**. This is where the probability crosses **0.5 (50%)**. In machine learning, this is the **Decision Boundary**.
   - A company ranked **40** is predicted as "High Market Cap."
   - A company ranked **60** is predicted as "Not High Market Cap."
3. **The Rapid Drop-off:** Between Rank 30 and 70, a small change in rank leads to a huge change in probability. This suggests that in the dataset, there is a very clear "threshold" where companies stop being considered industry giants.

Svm Model:

- This code transitions from simple clustering to **Supervised Machine Learning**. You are using a **Support Vector Machine (SVM)** to see if the financial
- (Revenue and Market Cap) of a company can predict whether it is based in the **USA** or **China**.

1. Advanced Data Cleaning (Pre-processing)

- **Unit Conversion:** The clean_numeric function handles the "B" (Billions) and "T" (Trillions) suffixes, converting strings like "$500.2B" into actual floating-point numbers.
- **Normalization:** clean_country standardizes names (e.g., "United States" becomes "USA") so the model doesn't treat them as different entities.
- **Binary Filtering:** The narrow the scope to just two classes (USA vs. China), which is a classic **Binary Classification**.

2. Feature Scaling

Python

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Why this is vital:

SVMs are sensitive to the "scale" of numbers. If Revenue is in the billions but Price is in the millions, the model will ignore Price because the numbers are smaller. Scaling puts both features on a level playing field (usually between -3 and 3).

3. The Support Vector Machine (SVM)

The SVM's role is to find the **Optimal Hyperplane**—a "DMZ" or boundary line that separates the USA companies from the Chinese companies with the widest possible gap (the **Margin**).

- **kernel="linear", Support Vectors**: These are the "extreme" data points near the boundary. The code highlights them with large circles. These specific companies are the hardest for the model to classify, and they actually define where the line is drawn.

4. Visualizing the Decision Boundary

The plot at the end is the most important part. It visualizes the model's

- **Contour Fill**: The red and blue shaded areas show the "territories." If a new company falls in the red zone, the model predicts it's from China; blue zone, USA.
- **The Solid Line**: This is the decision boundary.
- **The Dashed Lines**: These represent the **Margin**. SVM tries to make this gap as wide as possible to ensure the model is robust and not just guessing.

Summary of the Output:

- **Accuracy:** Tells the percentage of companies the model correctly assigned to the right country.
- **Classification Report:** Gives the "Precision" and "Recall"—essential if one country has many more companies than the other, as it prevents the model from being biased.

Hyperparameter:

Key Principles

Hyperparameters can be classified into two main categories:

1. **Model Hyperparameters**: These define the structure of the model, such as the number of layers in a neural network or the number of clusters in a k-means algorithm.
2. **Algorithm Hyperparameters**: These influence the learning process, such as the learning rate in gradient descent or the batch size in stochastic gradient descent.

Google Colab Link:

https://colab.research.google.com/drive/1yBCUAHNEgGEgJtZGmxRbuFVvwh8-M4-1?usp=sharing